Benjamin Thérien

1-(514)-823-7115·benjtherien@gmail.com · Scholar ·Github· LinkedIn· Website ·US & Canadian dual Citizen

Ph.D. at MILA (graduating 2026), researching pre-training optimizers, HP transfer & cross-datacenter training.

Research Experience

Google Research, DiLoCo Team — Seattle, WA
Research Scientist Intern, Advised by Zachary Charles — June 2026 - Sept 2026

- Proposed and led research investigating scalable per-tensor hyperparameters for pre-training optimizers, **leading to 15-20% compute-efficiency gains relative to MuonH** across a scaling ladder of **150M-3B** transformers.
- Met with a Gemini Muon pre-training lead to discuss **landing our research findings in Gemini**.
- Improved (**6.25**×) MoE step time by landing expert parallelism, ragged A2A, and a fused up/gate GMM.
- Improved compile time (> 9×) for a 7B transformer with layer stacking and lax.scan.

Meta Superintelligence Labs, FAIR Optimization Team — New York, NY
Research Scientist Intern, Advised by Aaron Defazio — June 2025 - Feb 2026

- Diagnosed silent memory leaks and slow multi-node interconnect using TorchTitan + TorchFT. [18]
- Implemented all-to-all reduce-scatter-compatible compression mechanisms within Torch Fault Tolerant. [18]
- Proposed MuLoCo, achieving the **largest critical batch size** of any cross-datacenter training method. [18]
- Improved FLOP-matched worker scaling in DiLoCo by 2-3% final loss relative to data-parallel. [18]
- Studied scaling laws for 150M-3.1B finding MuLoCo achieves > 1.3× FLOP Eff. w.r.t AdamW. [18]
- Validated MuLoCo at **15B**, achieving a 8× **wall-clock speedup** over DP AdamW with better performance. [18]

MILA Ph.D. — Montreal, QC
Ph.D. Researcher advised by Dr. Irina Rish, and Dr. Eugene Belilovsky — May 2023 - present

- Consulted on architecture, data mixture, parallelism, optimizer hyperparameters, and communication savings for **Covenant-72B** decentralized training run. [17,19]
- Pre-trained **10B** LLMs across 276 nodes on a PowerPC cluster, requiring re-compiling all dependencies. [8]
- Optimized MFU **for 410M-34B LLMs** on AMD and Nvidia GPU clusters. [8]
- Massively parallelized(> 100× faster) the downloading and tokenization of datasets for LLM pre-training. [8,11,15]
- Derived μP for popular learned optimizer architectures, achieving a 7x lower loss for large tasks. [10]
- Designed PyLO, a CUDA-accelerated open-source library for learned optimizers in PyTorch. [16]

Capital One, LLM Pre-training Team — New York, NY
Applied Research Intern — June 2024 - August 2024

- In a controlled scientific study, I established the properties of decoder-only MoE transformers compared to dense transformers in a realistic continual pre-training setup. [11]
- Pre-trained multiple 2.3**B dropless MoE LLMs** for over 1T tokens on an A100 AWS Kurbernetes cluster. [11]
- Implemented the latest DeepSeek MoE architecture in GPT-NeoX + custom DeepSpeed launcher. [11,13]
- Implemented asynchronous logging of MoE routing decisions, resulting in a 2× pre-training speedup. [11,13]
- Acheived 3% final loss improvement by tuning the granularity of the MoE. [11,13]

WISE Lab, University of Waterloo — Waterloo, ON
Masters Researcher, advised by Dr. Krzysztof Czarnecki — sept 2021 - April 2023

- Designed scalable data extraction pipelines and training code for object re-identification from point clouds [5].
- Led the first large-scale 3D computer vision study on point-based object re-identification through its entire lifecycle: ideation, implementation, analysis, writing, and publication [5].

MILA *Research Intern* — Montreal, QC, May 2021 - Aug 2021

Computational Linguistics Lab, *Undergraduate Researcher* — Montreal, QC, Sep 2020 - Apr 2021

SWE Intern Experience

Morgan Stanley, *Software Developer Intern* — Montreal, QC, Jan 2020 - Apr 2020
Accedian Networks, *Software Developer Intern* — Montreal, QC, May 2019 - Aug 2019

Education

Université de Montréal — Montreal, QC
Ph.D. in Computer Science — Sep 2023 - Present

Waterloo University — Waterloo, ON
Master of Mathematics Computer Science, Thesis — Sep 2021 - April 2023

Concordia University — Montreal, QC
Bachelor of Computer Science, Co-op — Sep 2017 - May 2021

Selected Publications & Preprints

*Equal contribution, †Oral presenter, ‡Senior Author

[21] Xiaolong Huang, **Benjamin Thérien**, James Harrison, Eugene Belilovsky.
Efficient Long-Horizon Learning for Learned Optimization. ***In submission to*** *ICLR 2027.*

[20] Stefan Horoi, **Benjamin Thérien**, Guy Wolf, Eugene Belilovsky. Can Model Merging Improve Aggregation in DiLoCo?. ***Accepted to*** *NeurIPS 2026.*

[19] Joel Lidin, Amir Sarfi, Erfan Miahi, Quentin Anthony, Shivam Chauhan, Evangelos Pappas, **Benjamin Thérien**, Eugene Belilovsky, Samuel Dare. Covenant-72B: Pre-Training a 72B LLM with Trustless Peers Over-the-Internet. *Preprint.*

[18] **Benjamin Thérien**, Xiaolong Huang, Aaron Defazio, Irina Rish, Eugene Belilovsky.
MuLoCo: Muon is a Practical Inner Optimizer for DiLoCo. ***Accepted to*** *ICML 2026.*

[17] Amir Sarfi, **Benjamin Thérien**, Joel Lidin, Eugene Belilovsky.
Communication Efficient LLM Pre-training with SparseLoCo. ***In submission to*** *NeurIPS 2026.*

[16] Paul Janson*, **Benjamin Thérien***, Quentin Anthony, Xiaolong Huang, Abhinav Moudgil, Eugene Belilovsky.
PyLO: Towards Accessible Learned Optimizers in PyTorch. ***Accepted to*** *MLSys 2026*.

[15] **(Oral)** Vaibhav Singh, Paul Janson, Paria Mehrbod, Adam Ibrahim, Irina Rish, Eugene Belilovsky, **Benjamin Thérien**‡. Beyond Cosine Decay: On the Effectiveness of Infinite Learning Rate Schedule for Continual Pre-training. ***Accepted to*** *CoLLAs 2025.*

[13] Ashwinee Panda, Vatsal Baherwani, Zain Sarwar, **Benjamin Thérien**, Stephen Rawls, Sambit Sahu, Supriyo Chakraborty, Tom Goldstein. Dense Backpropagation Improves Routing for Sparsely-Gated Mixture-of-Experts. ***Accepted to*** *NeurIPS 2025.*

[11] **Benjamin Thérien**, Charles-Étienne Joseph, Zain Sarwar, Ashwinee Panda, Anirban Das, Shi-Xiong Zhang, Stephen Rawls, Sambit Sahu, Eugene Belilovsky, Irina Rish.
Continual Pre-training of MoEs: How Robust is your Router?. ***Accepted to*** *TMLR 2025.*

[10] **(Oral, top 5.21%) Benjamin Thérien**†, Charles-Étienne Joseph, Boris Knyazev, Edouard Oyallon, Irina Rish, and Eugene Belilovsky. μLO: Compute-Efficient Meta-Generalization of Learned Optimizers. ***Accepted to*** *NeurIPS 2024 MLOpt Workshop (oral);* ***Accepted to*** *ICLR 2026 (poster).*

[9] Charles-Étienne Joseph*, **Benjamin Thérien***, Abhinav Moudgil, Boris Knyazev, Eugene Belilovsky.
Can We Learn Communication-Efficient Optimizers?. ***Accepted to*** *TMLR 2025*

[8] Adam Ibrahim*, **Benjamin Thérien***, Kshitij Gupta*, Mats Leon Richter, Quentin Anthony, Eugene Belilovsky, Timothée Lesort, and Irina Rish. Simple and Scalable Strategies to Continually Pre-train Large Language Models. ***Accepted to*** *TMLR 2024.*

[5] **Benjamin Thérien**, Chengjie Huang, Adrian Chow, Krzysztof Czarnecki.
Object Re-Identification from Point Clouds. ***Accepted to*** *WACV 2024*

[2] **(Oral, top 4.2%)** Shanel Gauthier*, **Benjamin Thérien***†, Laurent Alsène-Racicot, Muawiz Chaudhary, Irina Rish, Eugene Belilovsky, Michael Eickenberg, Guy Wolf. Parametric Scattering Networks. ***Accepted to*** *CVPR 2022.*

Selected Code Repositories/Projects & Open Source Contributions

PyLO: Learned Optimization for PyTorch *PyTorch*
PyLO provides CUDA-accelerated PyTorch implementations of cutting-edge learned optimizers. [18]

MuLoCo-1 *PyTorch, Jax*
Open source optimizer implementation of single-worker MuLoCo in PyTorch and Jax [16].

μLearned Optimization *Jax*
Meta-training and meta-testing code for μLOs [10].

GPT-NeoX *Python, DeepSpeed, Megatron*
Contributed a model conversion script (HF to GPT-NeoX 2.0 format) along with various bug fixes.

Parameteric Scattering Networks *Python, Pytorch, Kymatio, Git*
Research code for *Parametric Scattering Networks*[2]. Reproduces the results from our paper.

Point Cloud ReID *Python, Pytorch, Pytorch Distributed, OpenCV, Git*
Research code for *Object Re-identification from Point Clouds* [5]. Includes datasets and pre-trained models.

Skills and Qualifications

Languages	English, French
Citizenship	USA, Canada
Programming (current):	Python, Git, LaTeX
Programming (past):	C/C++, SQL, Java
Deep Learning libraries:	PyTorch, Jax, Transformers, DeepSpeed, Megatron, GPT-NeoX, TorchTitan, TorchFT

Scholarships

FRQNT Ph.D. Scholarship Sep 2023 - Sep 2027
FRQNT Supplement May-Sep 2021
NSERC USRA May-Sep 2021